ARJUNA JEE 2027

Lecture -01 | Mathematics

SEQUENCE & SERIES

By- Sachin Jakhar Sir



Topics *to be covered*

A General Second Degree Equation

B Arithmetic Progression

C Question Practice

D



Question Practice

Bumper + Challenger Question

If α, β, γ are roots of cubic : $x^3 - 3x^2 - 4x + 1 = 0$, then find value of:

(i) $\Sigma\alpha = 3$ (ii) $\Sigma\alpha\beta = -4$ (iii) $\alpha\beta\gamma = -1$

(iv) $\Sigma\alpha^2 = 17$ (v) $\Sigma\alpha^3 = 60$ (vi) $\Sigma\frac{1}{\alpha^2} = 22$

(vii) $\Sigma\alpha^2\beta = -9$ (viii) $\Sigma\alpha^5 = 958$ (ix) $\Sigma\frac{1}{\alpha^3+1}$

(ix) $\frac{1}{\alpha^3+1} + \frac{1}{\beta^3+1} + \frac{1}{\gamma^3+1} =$

(ix) $\frac{1}{\alpha^3+1}+\frac{1}{\beta^3+1}+\frac{1}{\gamma^3+1}=\frac{(\beta^3+1)(\gamma^3+1)+(\alpha^3+1)(\gamma^3+1)+(\alpha^3+1)(\beta^3+1)}{(\alpha^3+1)(\beta^3+1)(\gamma^3+1)}$

$=\frac{\beta^3\gamma^3+\beta^3+\gamma^3+1+\alpha^3\gamma^3+\alpha^3+\gamma^3+1+\alpha^3\beta^3+\alpha^3+\beta^3+1}{(\alpha^3+1)(\beta^3+1)(\gamma^3+1)}$

$=\frac{2\underbrace{(\alpha^3+\beta^3+\gamma^3)}_{60}+3+(\beta^3\gamma^3+\alpha^3\gamma^3+\alpha^3\beta^3)}{1+\underbrace{(\alpha^3+\beta^3+\gamma^3)}_{60}+(\beta^3\gamma^3+\alpha^3\gamma^3+\alpha^3\beta^3)+\alpha^3\beta^3\gamma^3}$

$(\alpha\beta\gamma)^3=(-1)^3$

$=$ ____ $\rightarrow$ DIBY $-$ (X)

$$(\beta^3\gamma^3 + \alpha^3\gamma^3 + \alpha^3\beta^3) = a^3 + b^3 + c^3$$

$$= 3abc + (a+b+c)\left(a^2+b^2+c^2 - (ab+bc+ca)\right)$$

Say $\begin{cases} a = \alpha\beta \\ b = \beta\gamma \\ c = \gamma\alpha \end{cases}$

$$= 3\alpha^2\beta^2\gamma^2 + (\alpha\beta+\beta\gamma+\gamma\alpha)\left(\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 - (\alpha\beta\gamma(\beta+\gamma+\alpha))\right)$$

$$a^3+b^3+c^3 = 3abc + (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$$

$$ab+bc+ca = \alpha\beta^2\gamma + \alpha\beta\gamma^2 + \alpha^2\beta\gamma$$

$$= \alpha\beta\gamma(\beta+\gamma+\alpha)$$

Acidic Question

Given that α, β, γ are all real roots of the equation $x^3 - 2007x + 2002 = 0$, then the value of $\frac{\alpha-1}{\alpha+1} + \frac{\beta-1}{\beta+1} + \frac{\gamma-1}{\gamma+1}$ is equal to

$y = \frac{x-1}{x+1}$

$\rightarrow x = \ldots$

Acidic Question

Old - IIT Jee.

Given that α, β, γ are all real roots of the equation $x^3 - 2007x + 2002 = 0$, then the value of $\frac{\alpha-1}{\alpha+1} + \frac{\beta-1}{\beta+1} + \frac{\gamma-1}{\gamma+1}$ is equal to

Sum of roots of new cubic !!

$x^3 - 2007x + 2002 = 0$ — roots α, β, γ

$y = \frac{x-1}{x+1}$

$xy + y = x - 1$

$y + 1 = x(1-y)$

$\frac{1+y}{1-y} = x$

HW

Cubic = 0 ??

roots: $\frac{\alpha-1}{\alpha+1}, \frac{\beta-1}{\beta+1}, \frac{\gamma-1}{\gamma+1}$

$\left(\frac{1+y}{1-y}\right)^3 - 2007\left(\frac{1+y}{1-y}\right) + 2002 = 0$

↳ Sum of Roots = ?

$\frac{(1+y)^3}{(1-y)^3} - 2007\frac{(1+y)}{(1-y)} + 2002 = 0$

$$\frac{(1+y)^3}{(1-y)^3} - 2007\frac{(1+y)}{(1-y)} + 2002 = 0$$

$$\frac{(1+y)^3 - 2007(1+y)(1-y)^2 + 2002(1-y)^3}{(1-y)^3} = 0$$

$$(1+y)^3 - 2007(1+y)(1-y)^2 + 2002(1-y)^3 = 0$$

$\frac{\alpha-1}{\alpha+1}$, $\frac{\beta-1}{\beta+1}$, $\frac{\gamma-1}{\gamma+1}$

$$1 + y^3 + 3y^2 + 3y - 2007(1 - y - y^2 + y^3) + 2002(1 - y^3 - 3y + 3y^2) = 0$$

$$1y^3 - 2007y^3 - 2002y^3 + 3y^2 + 2007y^2 + 3\times2002y^2 + \cdots y + \cdots = 0$$

$$(-4008)y^3 + (8016)y^2 + \cdots y + \cdots = 0$$

f_{nf} = Sum of roots

$$= -\frac{\text{coeff of } y^2}{\text{coeff of } y^3}$$

$$= +\frac{(8016)}{+4008} = 2$$

f_{nf}

Question

If α, β, γ are roots of $x^3 - x^2 - 1 = 0$ then find value of $\left(\frac{1+\alpha}{1-\alpha} + \frac{1+\beta}{1-\beta} + \frac{1+\gamma}{1-\gamma}\right)$ = _____.

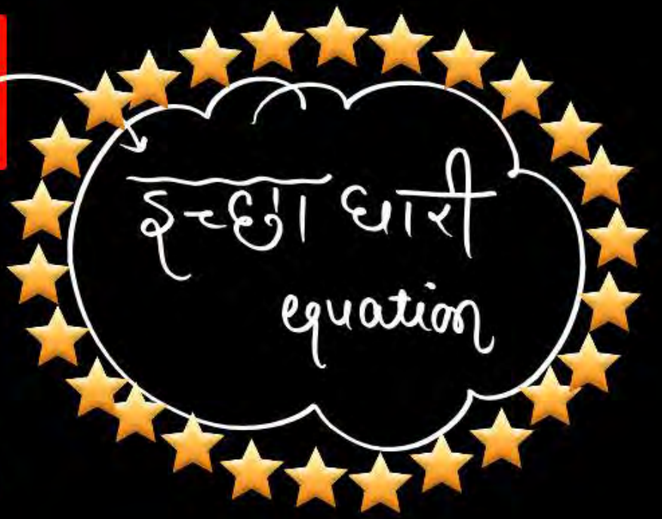
[Ans. : −5]

Q iBY - (Y)

SECOND DEGREE EQUATION

St., C, P, E, H

$$ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$$



multiplication of two St. lines (linear factors)

Condition

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

RAR

Question

Find value of k for which $12x^2 - 10xy + 2y^2 + 11x - 5y + k$ is a product of two linear factors. Hence find those factors.

- Part-01

$ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$

$a = 12$, $b = 2$, $2h = -10$, $2g = 11$, $2f = -5$, $k = c$

$h = -5$. $g = \frac{11}{2}$ $f = -\frac{5}{2}$

a, b, c & $f g h$

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$12 \times 2 \times k + 2\left(\frac{-5}{2}\right)\left(\frac{11}{2}\right)(-5) - 12 \times \left(\frac{25}{4}\right) - 2 \times \frac{121}{4_2} - k(25) = 0$$

$$24k + \frac{25 \times 11}{2} - 75 - \frac{121}{2} - 25k = 0$$

$$(2) \quad = 77 - 75 = k$$

Question

Find value of k for which $12x^2 - 10xy + 2y^2 + 11x - 5y + k$ is a product of two linear factors. Hence find those factors.

- Part-02

$$12x^2 - 10xy + 2y^2 + 11x - 5y + 2 = 0$$

Quad. in y or Quad. in x

$$2y^2 - 5(2x+1)y + 12x^2 + 11x + 2 = 0$$

$$12x^2 + (11 - 10y)x + (2y^2 - 5y + 2) = 0$$

Quad. in y or Quad. in x ⇓

$2y^2 - 5(2x+1)y + 12x^2 + 11x + 2 = 0$

$12x^2 + (11 - 10y)x + 2y^2 - 5y + 2 = 0$

new. (Q iBY-Z)

$$y = \frac{-(-5(2x+1)) \pm \sqrt{(-5(2x+1))^2 - 4 \times 2 (12x^2+11x+2)}}{2 \times 2}$$

$$y = \frac{-(-5(2x+1)) \pm \sqrt{(2x+3)^2}}{4} =$$

$(-5(2x+1))^2 - 4 \times 2 (12x^2+11x+2)$

$25(4x^2+4x+1) - 96x^2 - 88x - 16$

$100x^2 + 100x + 25 - 96x^2 - 88x - 16$

$4x^2 + 12x + 9 = (2x+3)^2$

$$y = \frac{-(-5(2x+1)) \pm \sqrt{(2x+3)^2}}{4} = \frac{10x+5 \pm (2x+3)}{4}$$

$+$

$$y = \frac{10x+5+2x+3}{4}$$

$$y = \frac{12x+8}{4}$$

$$\boxed{y = 3x+2}$$

$-$

$$y = \frac{10x+5-2x-3}{4}$$

$$y = \frac{8x+2}{4}$$

$$\boxed{y = 2x+\frac{1}{2}}$$

Question

If second degree equation in x & y :

$$2x^2 - y^2 + xy - 3x + 9y + k = 0$$

Can be resolved as product of two linear factors, then find k, also find those linear factors.

Ans: $k = -20$

Factors: $x + y - 4 = 0$ & $2x - y + 5 = 0$

Q IBY - (P)



SECOND DEGREE EQUATION

$$ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$$

Condition that it can be resolved as product of two linear factors is :

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$ax^2 + 2(hy+g)x + (by^2 + 2fy + c) = 0.$$

Proof:

$$ax^2 + 2(hy+g)x + (by^2+2fy+c) = 0$$ ← Quad. in x

$$x' = \frac{-2(hy'+g) \pm \sqrt{(2(hy+g))^2 - 4a(by^2+2fy+c)}}{2a}$$

- x' → linear factor x', y'
- $(by^2+2fy+c)$ → Quad. in y ⇓ Perfect square ⇓ Iss Quad. ka $D = 0$

Cond^n → $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

Quad. dec - 14

Recorded.

- Ques^{n}. on 2° eq^{n}. + PYQ's.
- Equations Reducible to Quad.



max. formula based.

easy but

* multiple concepts

Most Imp. Chapter of class- XI.

2 Ques^{n} - Mains

1 Ques^{n} - Adv.



INTRODUCTION

A.P., G.P., H.P., AGP, Misc.

"Sequence"

ek pattern ko follow करे

"Progression" ✓

T_n लिख सकें

"Series"

Terms of sequence with + or − sign

$T_n = n$ # 1, 2, 3, 4, ... → N. S ✓, P ✓

$T_n = n^2$ # 1, 4, 9, 16, 25 ... → squares of N. S ✓, P ✓

2, 3, 5, 7, 11, 13 ... → Prime No. S ✓, P ✗

$\frac{1+1}{2}, \frac{1+2}{2}, \frac{1+3}{2}, \frac{1+4}{2}, \frac{1+5}{2}$ .... $\left\langle \frac{n+1}{2} \right\rangle$ $T_n = \frac{n+1}{2}$

$S = \frac{1+1}{2} + \frac{1+2}{2} + \frac{1+3}{2} + ...$



ARITHMETIC PROGRESSION (A.P.)

Class-X OP in Chat!!

ex:- 1, 3, 5, 7, 9..... → A.P. ✓

$a_1, a_2, a_3, a_4, \ldots \rightarrow$ A.P.

$$a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = \ldots$$

A.P.

$$a,\ a+d,\ a+2d,\ a+3d,\ \ldots\ldots \rightarrow \text{A.P.}$$

First term $= T_1 = a$.

Common difference $= d$

$a,\ a+d,\ a+2d,\ a+3d,\ \ldots\ldots \rightarrow$ A.P.

First term $= T_1 = a$.

Common difference $= d$.

$T_2 = a+d$

$T_3 = a+2d$

$T_4 = a+3d$

$\vdots$

\# n^{th} Term $= T_n = a + (n-1)\,d$

$$S = a + (a+d) + (a+2d) + (a+3d) + \cdots + (a+(n-1)d)$$

$$S = (a+(n-1)d) + (a+(n-2)d) + (a+(n-3)d) + \cdots + a$$

add

$$2S = (2a+(n-1)d) + (2a+(n-1)d) + (2a+(n-1)d) + \cdots + (2a+(n-1)d)$$

Sum of first n Terms $= S_n = \frac{n}{2}(2a+(n-1)d) = \frac{n}{2}(1^{st} + last)$

57%

Question

If sequence $< a_n >$ constitute an A.P. and $a_3 + a_5 - a_2 = 13, a_1 + a_7 = 23$.
Then a_{10} = _____.

A. $\frac{37}{4}$

B. $\frac{3}{4}$

C. 16

D. 18

I

a, d

cd.

$a_3 = a+2d$

$a_5 = a+(5-1)d$

$= a+4d$

$a+2d + a+4d - (a+d) = 13$

$a+5d = 13$

$a + a+6d = 23$

$2x$ $2a+6d = 23$

$2a+10d = 26$

$-4d = -3$

$\# d = \frac{3}{4}$

$a = 13 - 5\times\frac{3}{4}$

$a = \frac{52-15}{4} = \frac{37}{4}$

$a_{10} = a+9d$

$= \frac{37}{4} + 9\times\frac{3}{4} = \frac{37+27}{4} = \frac{64}{4} = 16$

2021 – "Mains"

82%

Question

For an A.P. $S_{10} = 530$ & $S_5 = 140$, then value of $S_{20} - S_6$ is _____.

(Sum of first 10 Terms. / Sum of first 5 Terms.)

A. 1682

B. 1862

C. 1268

D. 6182

A.P. → a & d.

$$S_{10} = \frac{10}{2}\{2a + (10-1)d\} = 530 \quad -(1) \rightarrow a + \frac{9d}{2} = 53$$

$$S_5 = \frac{5}{2}\{2a + (5-1)d\} = 140 \quad -(2) \rightarrow a + 2d = 28$$

$$\frac{5d}{2} = 25$$

$$d = 10$$

$$a = 8$$

$$S_{20} = \frac{20}{2}\{2\times 8 + 19\times 10\}$$

$$S_6 = \frac{6}{2}\{2\times 8 + 5\times 10\}$$

Question

For an A.P., $S_p = q$ and $S_q = p$, then S_{p+q} = _____.

A $p + q$

B $p - q$

C $q - p$

D $-p - q$

a, d

$S_p = \frac{p}{2}\{2a + (p-1)d\} = q$

$S_q = \frac{q}{2}\{2a + (q-1)d\} = p$

$S_{p+q} = \left(\frac{p+q}{2}\right)\{2a + (p+q-1)d\}$

$2a + (p-1)d = \frac{2q}{p} \rightarrow 2a = \frac{2q}{p} + (p-1)\frac{2(p+q)}{pq}$

$2a + (q-1)d = \frac{2p}{q}$

$(-)$

$(p-1-q+1)d = 2\left(\frac{q}{p} - \frac{p}{q}\right) = 2\left(\frac{q^2-p^2}{pq}\right) = \frac{2(q-p)(q+p)}{pq} = (p-q)d$

$d = \frac{-2(p+q)}{pq}$

Question

For an A.P., $S_p = q$ and $S_q = p$, then S_{p+q} = _____.

- X^{th} Ques.
- Mains
- New.

A $p + q$

B $p - q$

C $q - p$

D $-p - q$

CHALLENGER QUESTIONS

2-days new.

Acidic Question

If the equations $x^3 + x^2 - 4x = 4$ and $x^2 + px + 2p = 0$ $(p \in R)$ have two roots common, then the value of p is

DIBY-A

A -2

B -1

C 1

D 3

Acidic Question

The values of k, for which the equation $x^2 + 2(k-1)x + k + 5 = 0$ posses atleast one positive root, are

DIBY-B

A $[4, \infty)$

B $(-\infty, -1) \cup [4, \infty)$

C $[-1, 4]$

D $(-\infty, -1]$

Challenger Question

If the roots of the equation $x^2 + (p+1)x + 2q - q^2 + 3 = 0$ are real and unequal $\forall\ p \in R$, then find the minimum integral value of $(q^2 - 2q)$.

DIBY-C

Challenger Question [JEE Mains-2021]

If α and β are the distinct roots of the equation $x^2 + (3)^{1/4}x + 3^{1/2} = 0$, then the value of $\alpha^{96}(\alpha^{12} - 1) + \beta^{96}(\beta^{12} - 1)$ is equal to

DIBY-D

A 56×3^{25}

B 56×3^{24}

C 52×3^{24}

D 25×3^{25}

Question

If α, β and γ are roots of $3x^3 - 4x^2 - 3x + 2 = 0$ and

$$(\alpha^5 + \beta^5 + \gamma^5) - (\alpha^3 + \beta^3 + \gamma^3) = \frac{2}{m}\left(2(\alpha^4 + \beta^4 + \gamma^4) - (\alpha^2 + \beta^2 + \gamma^2)\right)$$

Then value of m is

DIBY-E

Question

Let α, β, γ be roots of $x^3 + 2x^2 - 4x + 5 = 0$ find value of $\frac{(\alpha^3+5)(\beta^3+5)(\gamma^3+5)}{13\alpha\beta\gamma}$.

DIBY-F

Challenger Question [JEE Mains-2024]

Let $x_1, x_2, x_3, x_4,$ be the solution of the equation $4x^4 + 8x^3 - 17x^2 - 12x + 9 = 0$ and $(4 + x_1^2)(4 + x_2^2)(4 + x_3^2)(4 + x_4^2) = \frac{125}{16}m.$ Then the value of m is _____.

DIBY-G

M-1 RRT | M-2 i wala method | M-3 Transformation

Critical Thinking Question (CTQ)

Let α, β, γ be distinct real number and $f(x)$ is a quadratic polynomial such that

$$f(2)\alpha^2 + f(3)\alpha + f(4) = 4\alpha^2 + 4\alpha + 8$$

$$f(2)\beta^2 + f(3)\beta + f(4) = 4\beta^2 + 4\beta + 8$$

$$f(2)\gamma^2 + f(3)\gamma + f(4) = 4\gamma^2 + 4\gamma + 8$$

then find the value of $f(7)$.

DIBY-H

Question [JEE Mains-2023]

Let α, β, γ be the three roots of the equation $x^3 + bx + c = 0$.
If $\beta\gamma = 1 = -\alpha$, then $b^3 + 2c^3 - 3\alpha^3 - 6\beta^3 - 8\gamma^3$ is equal to

A 155/8

B 21

C 169/8

D 19

DIBY-I

Challenger Question [Multiple Correct]

If $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$ are roots of equation $x^5 - 5x^4 - 1 = 0$, then

DIBY-J

A $\sum_{r=1}^{r=5} \frac{1}{\alpha_r^4} = -\frac{1}{20}$

B $\sum_{r=1}^{r=5} \frac{1}{\alpha_r^4} = -20$

C $\prod_{r=1}^{r=5} \left(\frac{1}{\alpha_r^4} + 5\right)^5 = 1$

D $\prod_{r=1}^{r=5} \left(\frac{1}{\alpha_r^4} + 5\right)^3 = \frac{1}{5}$

$\Sigma \rightarrow$ Sum

$\Pi \rightarrow$ Prod.

Product/multip.

$\left(\frac{1}{\alpha_1^4} + 5\right)^3 \times \left(\frac{1}{\alpha_2^4} + 5\right)^3 \times \left(\frac{1}{\alpha_3^4} + 5\right)^3 \cdots$

Acidic Question

If α, β are roots of $ax^2 + bx + c = 0$, then the equation $ax^2 - bx(x-1) + c(x-1)^2 = 0$ has roots

DIBY-K

A $\frac{\alpha}{1-\alpha}, \frac{\beta}{1-\beta}$

B $\frac{1-\alpha}{\alpha}, \frac{1-\beta}{\beta}$

C $\frac{\alpha}{1+\alpha}, \frac{\beta}{1+\beta}$

D $\frac{1+\alpha}{\alpha}, \frac{1+\beta}{\beta}$

Question [JEE Mains-2021]

If $x^2 + 9y^2 - 4x + 3 = 0, x, y \in R$, then x and y respectively lie in the intervals.

DIBY-L

A $[1, 3]$ and $[1, 3]$

B $[1, 3]$ and $\left[-\frac{1}{3}, \frac{1}{3}\right]$

C $\left[-\frac{1}{3}, \frac{1}{3}\right]$ and $[1, 3]$

D $\left[-\frac{1}{3}, \frac{1}{3}\right]$ and $\left[-\frac{1}{3}, \frac{1}{3}\right]$

Critical Thinking Question (CTQ)

DIBY-M

a, b, c, d are real numbers such that

$(a+1)(b+1)(c+1)(d+1) = 1$

$(a+2)(b+2)(c+2)(d+2) = 2$

$(a+3)(b+3)(c+3)(d+3) = 3$

$(a+4)(b+4)(c+4)(d+4) = 4$

Find $(a+5)(b+5)(c+5)(d+5)$.

ANSWERS (A TO M)

DIBY-A : B
DIBY-B : D
DIBY-C : 4
DIBY-D : C
DIBY-E : 3
DIBY-F : 8
DIBY-G : 221
DIBY-H : 44
DIBY-I : D
DIBY-J : B,C
DIBY-K : C
DIBY-L : B
DIBY-M : 29

It is never about end result

It's always about Journey

Thank You

#futureIITians